DISCRETE MATHEMATICS · SEMESTER PROJECT

Graph Theory in Streaming Recommendations

Semester VII · Division B · Batch B2

SUBMITTED BY

C039 C043 C044 C046

1 CONCEPT

Concept from Discrete Mathematics

Graph Theory

Degree, Connectivity, Path, Cycle, Eulerian and Hamiltonian Graphs, Planar Graphs, and Graph Coloring.

Every idea used in this project comes directly from these seven topics. The recommendation problem is treated as a question about a graph, and each topic answers a different part of that question, from how popular a piece of content is, to how a user's watch queue should be ordered, to how the whole system is drawn and colored so it stays readable and varied.

2 OVERVIEW

Project Overview

> *This project looks at how streaming platforms decide what to show a user next, and treats that as a graph problem.*

The platform's users and content are represented as a graph, and each topic from the course is used to solve one specific part of the recommendation problem. Degree and connectivity are used to see which content is actually popular and whether users naturally form groups based on similar taste. They also help check whether the recommendation system can reach every group of users, or whether some are left out.

Path length is used to measure how relevant a piece of content is to a user, where a shorter path means a closer match. Cycles are used to spot a common problem, where a user keeps getting shown the same narrow set of content instead of anything new.

Eulerian and Hamiltonian graphs are used to build two different viewing experiences. An Eulerian circuit is used to create a playlist that moves smoothly between genres, and a Hamiltonian path is used to build a single, ordered watch list for one user. Planar graphs and graph coloring handle the final part of the project. A planar layout keeps the recommendation graph easy to read without lines crossing over each other, and graph coloring is used to make sure a user is not shown several very similar items in a row.

A small sample dataset of users and content is used throughout the project, so that each of these ideas can be shown working on the same example by the end.

3 TEAM

Team and Concept Ownership

ROLL NO.	CONCEPT OWNED
C039	Degree and Connectivity
C043	Path and Cycle
C044	Eulerian and Hamiltonian Graphs
C046	Planar Graphs and Graph Coloring

4 WEEK ONE

01

Foundations

ROLL NO.	TASK	DELIVERABLE
C039	Build the sample dataset of users and content items, and calculate the degree of every node in the graph.	Sample dataset and a degree table for each node.
C043	Define path and cycle for the graph using the sample dataset, and mark out a few real examples from it.	Written definitions with two or three worked examples of paths and cycles.
C044	Check whether the sample graph meets the conditions for an Eulerian or Hamiltonian graph, and note any changes needed if it does not.	A short note on which conditions are met, and what adjustments the dataset needs.
C046	Write the introduction and problem statement, explaining why graph theory fits this problem.	Introduction and problem statement, ready to place in the final report.

5 WEEK TWO

02

Core Work

ROLL NO.	TASK	DELIVERABLE
C039	Find the connected components in the graph, and use degree values to rank content by popularity.	List of connected components and a popularity ranking with reasoning.
C043	Calculate the shortest path from a sample user to every item, and use path length as a relevance score. Identify any cycles and explain what they mean for repeated recommendations.	Shortest path table with relevance scores, and a short explanation of any cycle found.
C044	Build a smaller genre transition graph, and work out an Eulerian circuit for a playlist and a Hamiltonian path for a single user's watch order.	Genre transition diagram with the Eulerian circuit and Hamiltonian path marked out.
C046	Try to draw the recommendation graph as a planar graph, and apply graph coloring so that similar items are not placed next to each other.	Planar layout attempt, or an explanation if it is not possible, and a colored graph diagram.

6 WEEK THREE

Integration and Report

03

ROLL NO.	TASK	DELIVERABLE
C039	Put together the final degree and connectivity results, along with one summary diagram.	Final section write-up with diagram.
C043	Put together the final path and cycle results, with the worked example carried through.	Final section write-up with the worked example.
C044	Put together the final Eulerian and Hamiltonian results, including the example playlist and watch order.	Final section write-up with the playlist and watch order example.
C046	Assemble the full report and presentation, and check that every section is consistent before submission.	Complete report and presentation, ready for submission.